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SUPERSEDING
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dated 27 Feb 1980

MILITARY SPECIFICATION

FUZE SETTER: M36

This specification is approved for use by the USA Armaments Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detailed requirements for FUZE SETTER: M36 used to set function times in the M587, M724 and M744 general purpose artillery electronic time fuzes.

1.2 Description. The M36 fuze setter is a digital electronic device capable of setting time and point detonation functions into the M587/M724/M744 fuzes. The fuze setter can set function times from 0.2 to 199.9 seconds in 0.1-second increments. The setter can interrogate the fuze to determine the setting to the nearest hundredth of a second without disturbing the set time.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or requests for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	-- Ammunition and Special Weapons, General Specification for
MIL-F-48700	-- FUZE, ELECTRONIC TIME: M587

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover, NJ, 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1390

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MIL-F-48702 -- FUZE, ELECTRONIC TIME: M724
MIL-Q-50829 -- General Quality Assurance Provisions for Proximity Fuzes and Related Components
MIL-F-50831 -- FUZE, TRAINING, ELECTRONIC TIME: M744

STANDARDS

Military

MIL-STD-105 -- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 -- Marking for Shipment and Storage
MIL-STD-461 -- Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 -- Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810 -- Environmental Test Methods
MIL-STD-1460 -- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for

DRAWINGS

USA Harry Diamond Laboratories (FSCM 19202).

PL11711372 -- Parts List for FUZE SETTER: M36 and accessories
EDL11711372 -- Special Inspection Equipment for FUZE SETTER: M36
10990455 -- Interface Hybrid Circuit
10990466 -- MNOS Counter Memory
11711256 -- MOS Scaler/Logic and Overhead Safety Circuit
11711379 -- "GO" Gage Fuze Setter
11711390 -- "NO GO" Gage Fuze Setter
11711427 -- Precision Oscillator

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Department of Defense

Handbook H4-1 -- Federal Supply Code for Manufacturers (part 1)

(Request for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

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3. REQUIREMENTS

3.1 Samples. The contractor is responsible for all requirements for providing and testing of samples as prescribed herein. When this specification is assigned to a subcontractor, it shall be the prime contractor's responsibility to insure that sample requirements are met. The prime contractor shall notify the Government at the time of assignment to a subcontractor whether the testing will be performed by the prime or by the subcontractor, as well as the method of control by the prime contractor.

3.1.1 First article approval sample. Prior to the start of regular production, except when production on a new contract at the same facility follows within 90 days production of acceptable material under this specification, the contractor shall manufacture and submit a first article approval sample of 20 fuze setter assemblies using as far as possible the methods and processes proposed for quantity production. When processes differ from the production processes, they shall be basically equivalent and the difference shall be fully described in documentation submitted with the first article sample. The sample shall conform to the requirements of this specification and shall be serialized. Additional samples that may be required because of failure of the sample to meet the requirements of this specification shall be supplied by the contractor at his expense. Prior to approval of the first article approval sample, acquisition of parts and materials or initiation of production will be at the sole risk of the contractor.

3.1.2 Supplemental samples. At the Government's discretion, a supplemental sample may be required whenever there is a break in production continuity of 90 days or more, or a change in design, material, or process that may affect safety, operability, reliability, or interchangeability as defined by the Requirements and Classification of Defects. In all cases, such changes will require prior approval of the procuring activity.

3.1.3 Comparison sample. One fuze setter assembly selected at random from each accepted lot (not from the acceptance sample) shall be shipped for evaluation within two working days of lot acceptance to the technical agency designated by the procuring activity. (See 6.1.3)

3.2 Construction. The device shall be constructed in accordance with the applicable drawing (PL11711372) to the extent prescribed herein.

3.2.1 Materials. Materials shall be those specified by the applicable drawings.

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3.2.2 Dimensions and technical notes.

3.2.2.1 Listed. Those dimensions and technical requirements listed in the Classification of Defects are mandatory.

3.2.2.2 Unlisted. The contractor may propose changes to characteristics shown on the drawings but not listed in the Classification of Defects, for the purpose of adapting the item to established manufacturing practices. Such proposals must be accompanied by evidence that the changes do not affect the function of the item and that all requirements will be met. If the Government confirms the contention of the contractor, the change will be approved for the duration of the contract. In case of dispute, the characteristics of the drawings shall apply. Approval of a change under provisions of this paragraph does not relieve the contractor from establishing and maintaining an adequate quality assurance program as elsewhere required. Provisions of this paragraph shall not be used to obtain approval for use of discrepant material (i.e., produced before approval is obtained); or for design changes, which should be requested in accordance with change provisions of the contract document.

Note: It is expected that the above changes will normally be accomplished prior to the construction of the first article approval sample and reflected therein.

3.2.2.3 Interchangeability. While the dimensions and tolerances shown in the drawings and specifications will generally insure satisfactory products, the contractor is notified that some selection or matching may be required to obtain combinations which provide proper function or economic use of components. The contractor is responsible for selecting combinations of tolerances within the specified limits which satisfy his process needs and also meet the specified performance and fit requirements.

3.3 Performance requirements. The performance requirements contained herein are mandatory.

3.3.1 Functional mode requirements.

3.3.1.1 Set time mode. With the mode switch in the SET TIME position (TI), the fuze setter shall be capable of setting a M587 fuze as specified in MIL-F-48700, a M724 fuze as specified in MIL-F-48702, and a M744 fuze as specified in MIL-F-50831 and indicating set times of 00.2 second through 199.9 seconds in 0.1-second increments when connected to a properly operating fuze. The setter shall have a setting accuracy of +0.08 and -0.06 second. The fuze setter shall indicate an E when the SECONDS

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switches are set to 000.0 or 000.1 second with the MODE switch set to the SET TIME position (TI) and the fuze setter connected to a properly operating fuze.

3.3.1.2 Interrogate mode. With the mode switch set to the INTERROGATE position(?), the fuze setter display shall indicate the fuze setting XXX.XX seconds when connected to a properly operating fuze that has been set to a set time of XXX.X and the interrogated display shall be within +0.08 and -0.06 second of the set time. If the fuze had been set to the point detonating mode the display shall indicate a P.

3.3.1.3 Point Detonation (PD) set mode. With the mode switch in the PD position (PD), the fuze setter shall set properly operating M587/M724/M744 fuzes to the PD mode and indicate a P.

3.3.2 Interface requirements.

3.3.2.1 J1, battery charge connector. Pin A shall be used to provide the +20 to +40V signal to charge the fuze setter internal battery. Pin B shall be used to provide the return (0V) for the +20 to +40V charge signal. Pin C shall be employed to check the fuze setter internal crystal oscillator frequency.

3.3.2.2 J2, fuze connector. Pin C and C' of J2 shall be the Ground contacts that interface with the outer ring of the fuze nose cone, and terminal C of the remote probe connector, J3. Pins B and B' shall be the MONITOR contacts that interface with the middle ring of the fuze nose cone, and terminal B of the remote probe connector, J3. Pin A of J2 shall be the -Vx contact that interfaces with the center pad of the fuze nose cone, and terminal A of the remote probe connector, J3. The -Vx contact shall be the last of the five contacts to connect to the fuze when placing the fuze setter or remote probe onto the fuze nose cone. The -Vx contact shall be the first of the five contacts to disconnect from the fuze when removing the fuze nose cone.

3.3.2.3 J3, remote probe connector. J3 shall have three pins of which pin A shall be connected to -Vx, pin B to monitor and pin C to ground.

3.3.3 Operating requirements.

3.3.3.1 Function time. An inert fuze that has been properly set by the setter shall be functioned to assure that the setter is setting the fuze to the indicated time ±0.1 second.

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3.3.3.2 Turn-on time. The fuze setter dc-to-dc converter shall turn on when the -Vx pin contacts the fuze for more than 225 milliseconds. The fuze setter converter shall not turn on when the -Vx pin contacts the fuze for less than 125 milliseconds.

3.3.3.3 Turn-off time. The fuze setter converter shall turn off if the -Vx pin disconnects from the fuze for more than 450 milliseconds. The fuze setter converter shall not turn off if the -Vx pin disconnects from the fuze for less than 250 milliseconds.

3.3.3.4 Fuze setter time base. The fuze setter time base shall have a frequency such that the period at the test point (pin C, J1) shall be 2500.00 ± 0.125 microseconds.

3.3.3.5 Loads. The fuze setter shall be capable of operating normally with a 3.0-k ohm load between the Monitor and -Vx interface pins, and with a 3.0-k ohm load between the Monitor and Ground interface pins. The fuze setter shall indicate an LE if the Monitor, -Vx, and Ground interface pins are electrically connected together.

3.3.3.6 Voltages. The positive polarizing voltages (+Vp) as measured at the monitor line shall be between 26.0 and 32.0 volts. The negative polarizing voltage (-Vp) shall be between -26.0 and -32.0 volts. The Vx voltage as measured at the Vx line shall be between -22.5 and -28.0 volts. Furthermore, the negative polarizing voltage shall be more negative than the Vx voltage by at least 2.5 volts.

3.3.3.7 Duration of polarizing pulses. The positive and negative polarizing pulses shall each be of a duration from 48.7 to 48.9 msec.

3.3.3.8 Pulse width circuit. The fuze setter shall be capable of setting a M587, M724, and a M744 fuze to any time from 0.4 sec to 180.6 sec if the fuze oscillator period is between 91 and 110 microseconds with a duty cycle of 50 ± 1 %. The fuze setter shall not set such a fuze to any time if the fuze oscillator has a period shorter than 65 microseconds or longer than 140 microseconds with a duty cycle of 50 ± 1 %.

3.3.3.9 Arming signal. The fuze setter shall indicate an error signal (E) if the arming signal from the fuze does not occur.

3.3.3.10 Fire signal. The fuze setter shall indicate an error signal (E) if the fire signal from the fuze does not occur.

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3.3.3.11 Charging circuit. The charging circuit shall control the charging current from 10 to 240 mA over the temperature range of -40°C to +63°C.

3.3.4 Remote probe requirements.

3.3.4.1 Remote probe. When the setter is used with the remote probe it shall meet the requirements indicated in paragraphs: 3.3.3.5, 3.3.3.6, 3.3.3.7.

3.3.5 Battery requirements (room temp.).

3.3.5.1 The setter battery shall be capable of being recharged at room temperature in 9 hours maximum from a 24-VDC source. The charge indicator shall appear on the display as a bar (-).

3.3.5.2 A setter with a fully charged battery at room temperature shall be capable of setting a minimum of 4000 properly operating fuzes before the low voltage indicator appears (L). After the low voltage indicator appears the setter shall be capable of setting a minimum of 90 properly operating fuzes.

3.3.6 Environmental requirements.

3.3.6.1 Low temperature. The fuze setter without remote probe shall meet all requirements of 3.3.1, 3.3.2, 3.3.3 and when used with a remote probe shall meet the requirements of 3.3.4 at a low temperature of -40°C after low temperature storage specified in MIL-STD-810, Method 502.1, Procedure I, except storage temperature shall be limited to -54°C. The setter battery shall charge at all temperatures down to -23°C.

3.3.6.1.1 The setter battery shall be capable of being recharged at -23°C in 67 hours maximum from a 20-VDC source. The charge indicator shall appear on the display as a bar (-).

3.3.6.1.2 A setter that has been conditioned at -40°C and has a battery that has been fully charged at -23°C shall be capable of setting a minimum of 1000 properly operating fuzes before the low voltage indicator appears. After the low voltage indicator appears the setter shall be capable of setting a minimum of 90 properly operating fuzes.

3.3.6.2 High temperature. The fuze setter without remote probe shall meet all requirements of 3.3.1, 3.3.2, 3.3.3 and when used with a remote probe shall meet the requirements of 3.3.4 at a high temperature of 63°C after high temperature storage specified in MIL-STD-810, Method 501.1, Procedure I. The battery shall charge at all temperatures up to 55°C.

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3.3.6.2.1 The setter battery shall be capable of being recharged at 55°C in 9 hours maximum from a 40-VDC source. The charge indicator shall appear on the display as a bar (-).

3.3.6.2.2 A setter that has been conditioned at 63°C and has a battery that has been fully charged at 55°C shall be capable of setting a minimum of 1000 properly operating fuzes before the low voltage indicator appears. After the low voltage signal appears the setter shall be capable of setting a minimum of 90 properly operating fuzes.

3.3.6.3 Electromagnetic interference. The fuze setter with the remote probe shall, while meeting all requirements of 3.3.1.1, meet limits of RE02 of MIL-STD-461. Except the test of RE02 shall be confined to only the radiated transient phase and shall not exceed the values shown in Figure 18 at the test distance specified in Table VIII. The fuze setter with the remote probe shall meet all the requirements of 3.3.1.1, while being subjected to environments RS01, RS02 and RS03 of MIL-STD-461.

3.3.6.4 Leakage (immersion). The fuze setter without remote probe shall meet all of the requirements of 3.3.1, 3.3.2, 3.3.3 and when used with a remote probe shall meet the requirements of 3.3.4 after being subjected to conditioning in accordance with MIL-STD-810, Method 512.1, Procedure I.

3.3.6.5 Dust (fine sand). The fuze setter without remote probe shall meet all of the requirements of 3.3.1, 3.3.2, 3.3.3 and when used with a remote probe shall meet the requirements of 3.3.4 after being subjected to the conditioning in accordance with MIL-STD-810, Method 510.1, Procedure I.

3.3.6.6 Shock. The fuze setter without remote probe shall meet all of the requirements of 3.3.1, 3.3.2, and 3.3.3 and when used with a remote probe shall meet requirements of 3.3.4 after being subjected to tests performed in accordance with MIL-STD-810, Method 516.2, Procedure II. In addition the setter and remote probe shall meet the same requirements after being subjected to the same test when the setter is in its carrying case with accessories.

3.3.6.7 Vibration. The fuze setter without remote probe shall meet all of the requirements of 3.3.1, 3.3.2, and 3.3.3 and when used with a remote probe shall meet requirements of 3.3.4 after being subjected to test performed in accordance with MIL-STD-810, Method 514.2, Procedure XI. In addition the setter and remote probe shall meet the same requirements after being subjected to the same test when the setter is in its carrying case with accessories.

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3.3.6.8 Humidity. The fuze setter without remote probe shall meet all of the requirements of 3.3.1, 3.3.2, 3.3.3 and when used with a remote probe shall meet the requirements of 3.3.4 during and after conditioning in accordance with MIL-STD-810, Method 507.1, Procedure II.

3.4 Workmanship. All parts shall be manufactured and finished in a thoroughly workmanlike manner to insure satisfactory functioning and durability. (See MIL-A-2550, Condition of Materials, Parts and Assemblies.)

3.4.1 Soldering. Soldering shall be in accordance with MIL-STD-1460.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Contractor quality assurance system. The contractor shall provide and maintain an adequate quality assurance system in compliance with MIL-Q-50829.

4.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification in compliance with MIL-Q-50829.

4.3 First article approval sample. The first article approval sample (FAAS) as specified in 3.1.1 shall be examined and tested in accordance with the FAAS and lot sample flow chart (Figure 1), operating tests (Table 1), Classification of Defects (4.4.5), and Material and Component Certification (4.4.7). The tests shall be performed by the contractor under the observation of the Government QA representative and/or a representative of the cognizant technical agency. The contractor shall supply complete data in reproducible form on the contractor tests and examinations of the FAAS units. The data and tested samples shall be delivered to the technical agency.

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4.3.1 Approval of first article sample. If the sample passes the criteria of specified examinations and tests, it shall be approved. If the sample fails in any of the examinations and tests, a failure analysis shall be conducted by the contractor under Government surveillance. The results of the tests and failure analysis of the units, together with the engineering analysis of the units, shall form a basis for corrective action. Depending upon the degree of corrective action deemed necessary by the Government, the FAAS may be:

a. Conditionally approved with the provisions that the recommended changes are made by the contractor prior to start of regular production or during production of the first lot as prescribed by the Government. Such portions of first article inspections needed to verify corrective action may be included in the acceptance requirements of the lot, at the Government's discretion.

b. Disapproved, and new samples required for the tests failed and for related tests which might be invalidated by the corrective action.

c. Disapproved, and a complete new sample required for approval.

In all cases, the contractor shall comply with any required changes for the duration of the contract.

4.3.1.1 Reinstitution of tests. Acceptance of the first article approval sample does not relieve the contractor from meeting all requirements of this specification throughout the contract. The Government reserves the right to independently verify all requirements by repeating first article approval sample tests or any portion thereof on any production lot. If such verification indicates failure to meet requirements previously verified by first article approval sample tests, the contractor will be required to take corrective action. If the requirements are not met on the first lot to which the correction can be applied, the Government reserves the right to institute tightened inspection on a lot basis, by adding to the normal lot acceptance inspections all or such parts of the first article approval sample tests needed to verify corrective action. Normal acceptance inspection will be resumed after two consecutive lots have satisfactorily passed the tightened conditions.

4.4 Acceptance inspection. Inspection shall be as specified in MIL-A-2550 and in this document. The contractor shall notify the Government QA representative prior to proceeding with inspection for Classification of Defects (4.4.5) and Operating Tests (4.4.6). Inspection may be made at a higher assembly level

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provided that equal quality assurance provisions are satisfied, but not at a lower level than specified.

4.4.1 Lot formation. A lot shall consist of no more than 500 devices produced at one manufacturing location under the following conditions: continuous production, stabilized methods and techniques, and the same revision of drawings and specifications except for those introduced without obsolescence.

4.4.2 Sampling. At the time a completed lot is presented to the Government for acceptance, the designated Government representative shall select from the lot a random sample of 20 devices to conduct required electrical tests for establishing the acceptability of the lot. Unless otherwise specified, MIL-STD-105 inspection level 1 applies.

4.4.3 Lot acceptance. Lots shall be accepted on the basis of Classification of Defects (4.4.5), Material and Component Certification (4.4.7), and Operating Tests (4.4.6) as indicated in lot sample test flow chart (Figure 1) and Table 1, except that tests B1, B2, B3, and B4 of Figure 1 shall be deleted.

4.4.4 Data recording. A written record shall be made of the results of all examination and tests performed as specified elsewhere in section 4 of this specification. The test data shall be recorded on 80 column standard data processing punch cards, or other computer compatible input medium approved by the Government. Data from each operating tests shall be recorded, but only the defect summary need be recorded in the inspection of the Classification of Defects. Data layout of the card shall be approved by the Government. The completed cards shall be forwarded to the designated Government agency within two working days after Government acceptance of the item or lot. If inspections are conducted on a sublot basis, the acceptance data for each sublot shall be so identified. (Engineering information, section 6).

4.4.4.1 Variables data. Variables data shall be recorded for all electrical tests (Table 1).

4.4.5 Classification of defects. The lot shall be 100 percent screened by using the classification of defects and applicable drawings.

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4.4.5.1 Parts and assemblies.

4.4.5.1.1 Fuze Setter: M36 (DWG 11711372).

Categories	Defects	Methods of Inspection
Critical	None defined	
Major		
101	Contact pins on setter and also on remote probe do not protrude sufficient length.	Go GAGE, P/N 11711379
102	Contact pins on setter and also on remote probe exceed maximum eccentricity.	No-Go GAGE, P/N 11711390
103	Battery Charge Connector damaged.	Visual
104	Remote Probe Connector damaged.	Visual
105	Setting switches damaged	Visual
106	Display damaged.	Visual
107	Maximum length of center contact pin on setter and on remote probe shall be .040 in. Maximum length of other pins shall be .060 in.	Standard Measuring Instrument
108	Remote probe damaged.	Visual
109	Remote probe cable damaged.	Visual
110	Charging cable damaged.	Visual
111	Carrying case and insert damaged.	Visual
112	Panel gasket improperly installed.	Visual DS-1
Minor		
201	Marking illegible or missing	Visual
202	Foreign matter in contact area.	Visual
203	Carrying case and setter handles inoperable.	Visual
204	Cleaning brush damaged.	Visual

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4.4.5.1.2 Defect standard.

DS-1 The edge of the panel gasket shall be flush with the surface of the housing or the panel. The gasket shall not be below or protrude from either surface by more than 1/32 inch.

4.4.6 Operating tests.

4.4.6.1 Lot operating tests. The sample shall be examined and tested by the contractor in accordance with the Classification of Defects (4.4.5), Component Certification (4.4.7), and Group A and B Inspections (4.4.6.1.1) as indicated in the FAAS and Lot Sample Flow Chart (Figure 1).

4.4.6.1.1 Group A and B inspection. Group A inspection shall consist of electrical tests (Table 1) at ambient temperatures, low temperatures (4.6.2.1) and high temperatures (4.6.2.2). Group B inspections shall consist of electrical tests after environmental conditioning prescribed in 4.6.2.3 through 4.6.2.8.

4.4.6.1.2 Disposition of lot samples. If the tested samples pass all the tests then all items except 11 through 15 of the sample of 20 items shall be returned to the lot. Any visual defects shall be repaired prior to returning setters to the lot. Within two days after completion of the acceptance testing, the remaining 5 items together with the required lot and acceptance data shall be forwarded to the technical agency designated by the procuring activity.

4.4.6.2 First article approval sample inspection. The first article approval sample submitted for qualification shall be subjected to the inspection for Classification of Defects (4.4.5) and Group A and B Inspections (4.4.6.1.1) as indicated in the FAAS and Lot Sample Flow Chart (Figure 1). The contractor shall perform these tests under the direction of the Government QA representative and/or a representative of the cognizant Government technical agency. However, the Government may elect to perform these tests at a Government laboratory.

4.4.7 Material and component certification. Prior to acceptance under this procurement, the contractor shall demonstrate by means of certifications or statement of findings that only materials and components conforming to the item specification and drawings have been used.

4.5 Test conditions and equipment.

4.5.1 Test conditions. Unless otherwise specified, all measurements are to be made at normal ambient room conditions of temperature, relative humidity and altitude. The following table

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of tolerances shall be applied where applicable to specified test conditions unless otherwise defined in the test procedures:

a.	Temperature in test chamber	± 2°C
b.	Relative humidity	± 5 percent
c.	Vibration amplitude	±10 percent
d.	Shock acceleration	±10 percent
e.	Physical measurements	± 5 percent

4.5.2 Inspection equipment. Inspection equipment shall be in accordance with the design level shown by the applicable EDL 11711372 and of sufficient accuracy and quality to permit performance of the required inspection. Gage and instrument setting shall be a function of this equipment, and independent of the operator. Equipment shall be marked to indicate the level of acceptance for each test. Test procedures and calibration procedures shall conform to MIL-Q-50829.

4.5.2.1 Tester. All the counters and the turn-off logic in the setter tester (Figure 2) shall each have an input impedance greater than 1 megohm and an input capacitance less than 100 picofarads. Equipment shall be capable of measurements as follows:

a. Voltages to the nearest 0.1 V.

b. Currents to the nearest 2 percent

c. Time interval to the nearest 10 ms for the turn-on and turn-off times and to the nearest 10 µs for the duration of -Vp and +Vp and to the nearest 1 ms for function time.

d. Period measurement to the nearest 10 ns. (1000 period averag

4.6 Test procedures.

4.6.1 Operating tests.

4.6.1.1 Electrical test. Electrical tests 1.1 through 1.41 shall be conducted as indicated in Table 1 with the appropriate test equipment indicated in Figure 2 and paragraph 4.5.2.1. If the fuze setter without remote probe or when tested with its accessories does not display the readings listed in Table 1 or test equipment does not register the values prescribed by 3.3.1, 3.3.2, 3.3.3, 3.3.4, and 3.3.5, the setter shall fail the test and be rejected. When a Maximum or Minimum limit is specified, the value is given in terms of absolute numbers.

4.6.1.2 Turn-on and turn-off times. The turn-on and turn-off times shall be measured with a counter as shown in Figure 2.

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4.6.1.3 Function time test. The function time shall be measured as shown in Figure 2.

4.6.1.4 Voltage test (+Vp, -Vp, -Vx). The +Vp, -Vp and -Vx voltages shall be measured with a voltmeter as shown in Figure 2. The voltmeter shall be capable of measuring voltages that have a duration of 45 to 55 msec.

4.6.1.5 Duration of +Vp and -Vp voltage pulses. The duration of the +Vp and -Vp voltage pulses shall be measured with a counter as shown in Figure 2.

4.6.1.6 Pulse width circuit tests. The pulse width circuit tests shall be performed using the square wave oscillator as shown in Figure 2.

4.6.1.7 Battery charging tests (room temp.). The number of sets that are available from a fully charged battery shall be measured with an events counter as shown in Figure 2. The battery shall be discharged by using a setter, that is set to T188.8, to set a properly operating fuze with a duty cycle of 1.98 seconds min on and 30.3 seconds max off. The setter shall pass the battery requirements 3.3.5.

4.6.1.8 Remote probe test. The setter, when used with a remote probe, shall be tested according to the following tests in Table 1: 1.22, 1.23, 1.24, 1.36 thru 1.41.

4.6.1.9 Arming signal. The arming signal shall be inhibited by closing switch S9 as shown in Figure 2.

4.6.1.10 Fire signal. The fire signal shall be inhibited by opening switch S10 as shown in Figure 2.

4.6.1.11 Charging circuit. The charging current shall be measured over the temperature range of -40°C to +63°C as shown in test 1.48 of Table I and Figure 2 with a charging voltage of 20 volts. The stabilization time at each temperature shall be 1.0 hour

4.6.2 Environmental tests.

4.6.2.1 Low temperature tests. The fuze setter without remote probe shall pass operating tests (4.6.1.1) and when used with a remote probe shall pass operating tests (4.6.1.8) under stabilized conditions specified in MIL-STD-810, Method 502.1, Procedure I, except storage temperature shall be limited to -54°C. The operating tests shall be conducted at temperatures limited to -40°C.

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4.6.2.1.1 Battery requirements (low temp.). The setter battery shall be tested according to 4.6.1.7 and shall pass the battery requirements of 3.3.6.1.1 and 3.3.6.1.2.

4.6.2.2 High temperature tests. The fuze setter without remote probe shall pass operating tests (4.6.1.1) and when used with a remote probe shall pass operating tests (4.6.1.8) under stabilized conditions specified in MIL-STD-810 Method 501.1, Procedure I. The operating tests shall be conducted at temperatures limited to +63°C.

4.6.2.2.1 Battery requirements (high temp.). The setter battery shall be tested according to 4.6.1.7 and shall pass the battery requirements 3.3.6.2.1 and 3.3.6.2.2.

4.6.2.3 Electromagnetic interference test. The fuze setter with remote probe shall pass electrical test 1.1 of Table 1 while being subjected to environments RS01, RS02, and RS03, MIL-STD-461. Also while performing electrical test 1.1 of Table 1, the emission level of the setter shall not exceed limits RE02 of MIL-STD-461 when measured by method RE02 of MIL-STD-462. Except that test of RE02 in MIL-STD-461 shall be confined to only the radiated transient phase and shall not exceed the values shown in Figure 18 at the test distances specified in Table VIII.

4.6.2.4 Leakage (immersion). The fuze setter with remote probe shall be subjected to conditioning in accordance with MIL-STD-810, Method 512.1, Procedure I, and thereafter pass operating tests (4.6.1.1) and Classification of Defects (4.4.5.1.1) and when used with a remote probe shall pass operating tests 4.6.1.8.

4.6.2.5 Dust (fine sand). The fuze setter with remote probe shall be subjected to conditioning in accordance with MIL-STD-810, Method 510.1, Procedure I, and thereafter pass operation tests (4.6.1.1) and Classification of Defects (4.4.5.1.1) and when used with a remote probe shall pass operating tests 4.6.1.8.

4.6.2.6 Shock. The fuze setter with remote probe and also when in a carrying case with accessories shall be subjected to conditioning in accordance with MIL-STD-810, Method 516.2, Procedure II and thereafter pass operating tests (4.6.1.1), Classification of Defects (4.4.5.1.1) and when used with a remote probe shall pass operating tests (4.6.1.8).

4.6.2.7 Vibration. The fuze setter with remote probe and also when in a carrying case with accessories shall be subjected to conditioning in accordance with MIL-STD-810, Method 514.2, Procedure XI, and thereafter pass operating tests (4.6.1.1), Classification of Defects (4.4.5.1.1) and when used with a remote probe shall pass operating tests (4.6.1.8).

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4.6.2.8 Humidity. The fuze setter with remote probe shall be subjected to conditioning in accordance with MIL-STD-810, Method 507.1, Procedure II with temperature limited to 63°C and pass all operating tests (4.6.1.1) and Classification of Defects (4.4.5.1.1) and when used with a remote probe shall pass operating tests (4.6.1.8).

5. PREPARATION FOR DELIVERY

5.1 Cleaning, Preservation and Packaging. The cleaning, preservation and packaging of items for protection against corrosion, physical damage and other forms of deterioration during handling, shipment and storage shall be in accordance with applicable drawings.

5.2 Metal Oxide Semiconductor (MOS) device handling. During interplant and intraplant shipment, normal MOS handling and shipping shall be observed for all of the integrated circuits and any of the assemblies of which they are a part. Normal handling consists of preventing static charge from being applied to the devices.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Items to be included in the procurement documents.

6.1.1 Documents. Procurement documents should specify the title, date and number of this specification.

6.1.2 Samples. The number of units manufactured must be increased over the number ordered so as to include the samples as required in 3.1 and 4.4.2.

6.1.3 Government testing of comparison sample. These samples are subject to retesting by the Government to insure that first article approval requirements continue to be met. Failure of these samples will not affect the acceptance of the lot, but may indicate future mandatory action.

6.1.4 Lead time. The testing of the First Article Approval Sample will require a maximum of 40 working days. If there are failures, additional time will be required.

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6.1.5 Data requirements. The following data are required by this specification to be furnished to the Government and should be entered on DD form 1423 for each contrac:

First Article Test Results (4.3)
Examination and Test Records on Data Processing Cards (4.4.4)
MIL-Q-50829, Para 3.6.1 -- Contractor-furnished designs

6.2 Advisory documents. The following documents are not part of this specification, but may provide useful information to the contractor. Copies are available for reference at the Harry Diamond Laboratories, 2800 Powder Mill Road, Adelphi, MD. 20783.

6.2.1 Manufacturing Process Report or Description of Manufacture.

6.2.2 Quality Assurance System Documents Obtained from Previous Contracts.

6.2.3 Flow Chart M36 Fuze Setter (DWG 11711661).

6.3 Engineering information. Upon request, copies of the data collected in the acceptance of a lot will be made available to the Government representative. Additional data or data requirements, when needed for engineering purposes, will be specifically defined by the contract.

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TABLE 1. ELECTRICAL TESTS

Test No.	Test	Setting Switches	Display Readout	Notes/Req. Para.
1.1	Set Time	T188.8	T188.8	3.3.1.1
1.2	Interrogate	?---.-[1]	188.74 to 188.88	3.3.1.2
1.3	Set Time	T099.9	99.9	3.3.1.1
1.4	Set Time	T100.0	100.0	3.3.1.1
1.5	Set Time	T011.1	11.1	3.3.1.1
1.6	Set Time	T122.2	122.2	3.3.1.1
1.7	Set Time	T033.3	33.3	3.3.1.1
1.8	Set Time	T144.4	144.4	3.3.1.1
1.9	SeT Time	T055.5	55.5	3.3.1.1
1.10	Set Time	T166.6	166.6	3.3.1.1
1.11	Set Time	T077.7	77.7	3.3.1.1
1.12	Set Time	T000.0	E	3.3.1.1
1.13	Interrogate	?---.-	P	3.3.1.2
1.14	Set Time	T000.1	E	3.3.1.1
1.15	Interrogate	?---.-	P	3.3.1.2
1.16	Set Time	T000.2	00.2	3.3.1.1
1.17	Interrogate	?---.-	00.17 to 00.24	3.3.1.2
1.18	Set Time	T199.9	199.9	3.3.1.2
1.19	Interrogate	?---.-	199.84 to 199.98	3.3.1.2
1.20	PD Mode	P---.-	P	3.3.1.3
1.21	Interrogate	?---.-	P	3.3.1.2
1.22	Set Time	T188.8	188.8	Connect 3.0kΩ±5%, between MON/Vx 3.3.3.5
1.23	Set Time	T188.8	188.8	Connect 3.0kΩ±5% between MON/GND 3.3.3.5
1.24	Set Time	T188.8	188.8	Connect 3.0kΩ±5% between MON/Vx MON/GND 3.3.3.5
1.25	Set Time	T188.8	LE	Short Circuit, MON/Vx/GND 3.3.3.5
1.26	Set Time	T188.8	188.8	3.3.1.1
1.27	Set Time	T005.6	05.6	3.3.1.1
1.28	Interrogate	?---.-	05.54 to 05.68	3.3.1.2
1.29	Function Time	-	-	Test as indicated in Figure 2.Function time shall be interrogate time ± 0.020 s.
1.30	Oscillator Square Wave at:			
	91 µs±1 µs	T001.5	01.5	3.3.3.8
	65 µs±1 µs	T001.5	E	3.3.3.8
	110 µs±1 µs	T001.5	01.5	3.3.3.8
	140 µs±1 µs	T001.5	E	3.3.3.8
1.31	Arming Signal	T188.8	E	3.3.3.9
1.32	Fire Signal	T188.8	E	3.3.3.10

[1] The time setting switches can be in any position during interrogate.

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TABLE 1. ELECTRICAL TESTS (CONT'D)

Test NO.	Test	Minimum	Maximum	Unit	Req. Para.
1.33	Turn-On Time	125	225	ms	3.3.3.2
1.34	Turn-Off Time	250	450	ms	3.3.3.3
1.35	0.4 kHz Test Frequency Period	2499.875	2500.125	µs	3.3.3.4
1.36	Set to 1.5 sec, verify set measure +Vp at:				
	ambient temperature	28.0	30.0	V	3.3.3.6
	+63°C	28.5	32.0	V	3.3.3.6
	-40°C	26.0	28.5	V	3.3.3.6
1.37	Measure -Vp at:				
	ambient temperature	-28.0	-30.0	V	3.3.3.6
	+63°C	-28.5	-32.0	V	3.3.3.6
	-40°C	-26.0	-28.5	V	3.3.3.6
1.38	Measure -Vx at:				
	ambient temperature	-25.0	-26.8	V	3.3.3.6
	+63°C	-26.0	-28.0	V	3.3.3.6
	-40°C	-22.5	-25.0	V	3.3.3.6
1.39	Calculate -Vp-(-Vx) at:				
	ambient temperature	-2.5		V	3.3.3.6
	+63°C	-2.5		V	3.3.3.6
	-40°C	-2.5		V	3.3.3.6
1.40	Measure duration of +Vp at:				
	ambient temperature	48.7	48.9	ms	3.3.3.7
	+63°C	48.7	48.9	ms	3.3.3.7
	-40°C	48.7	48.9	ms	3.3.3.7
1.41	Measure duration of -Vp at:				
	ambient temperature	48.7	48.9	ms	3.3.3.7
	+63°C	48.7	48.9	ms	3.3.3.7
	-40°C	48.7	48.9	ms	3.3.3.7

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TABLE 1. ELECTRICAL TESTS (CONT'D)

Test No.	Test	Minimum	Maximum	Unit	Req. Para.
1.42[z]	Number of fuze settings, after charge at -23°C for 67 hours at 20 volts, before Low Voltage indicator appears. Observe charging indicator (-).	1000			3.3.6.1.1 3.3.6.1.2
1.43	Number of fuze settings after low voltage indicator appears.	90			3.3.6.1.2
1.44[z]	Number of fuze settings after charge at room temperature for 9 hours at 24 volts, before low voltage indicator appears. Observe charging indicator (-).	4000			3.3.5.1 3.3.5.2
1.45	Number of fuze settings after low voltage indicator appears.	90			3.3.5.2
1.46[z]	Number of fuze settings after charge at 55°C for 9 hours at 40 volts, before low voltage indicator appears. Observe charging indicator (-).	1000			3.3.6.2.1 3.3.6.2.2
1.47	Number of fuze settings after low voltage indicator appears.	90			3.3.6.2.2
1.48	Charging current at the following temperatures:				3.3.3.11
	°C				
	-40	6.7	34.0	mA	
	-23	6.7	53.3	mA	
	0	56.5	166.3	mA	
	20	171.5	324.6	mA	
	40	218	324.6	mA	
	55	218	324.6	mA	
	63		9	mA	

[z] Note: Before the start of these tests, the Setter battery shall be discharged at room temperature to where the Setter display reads (L) only.

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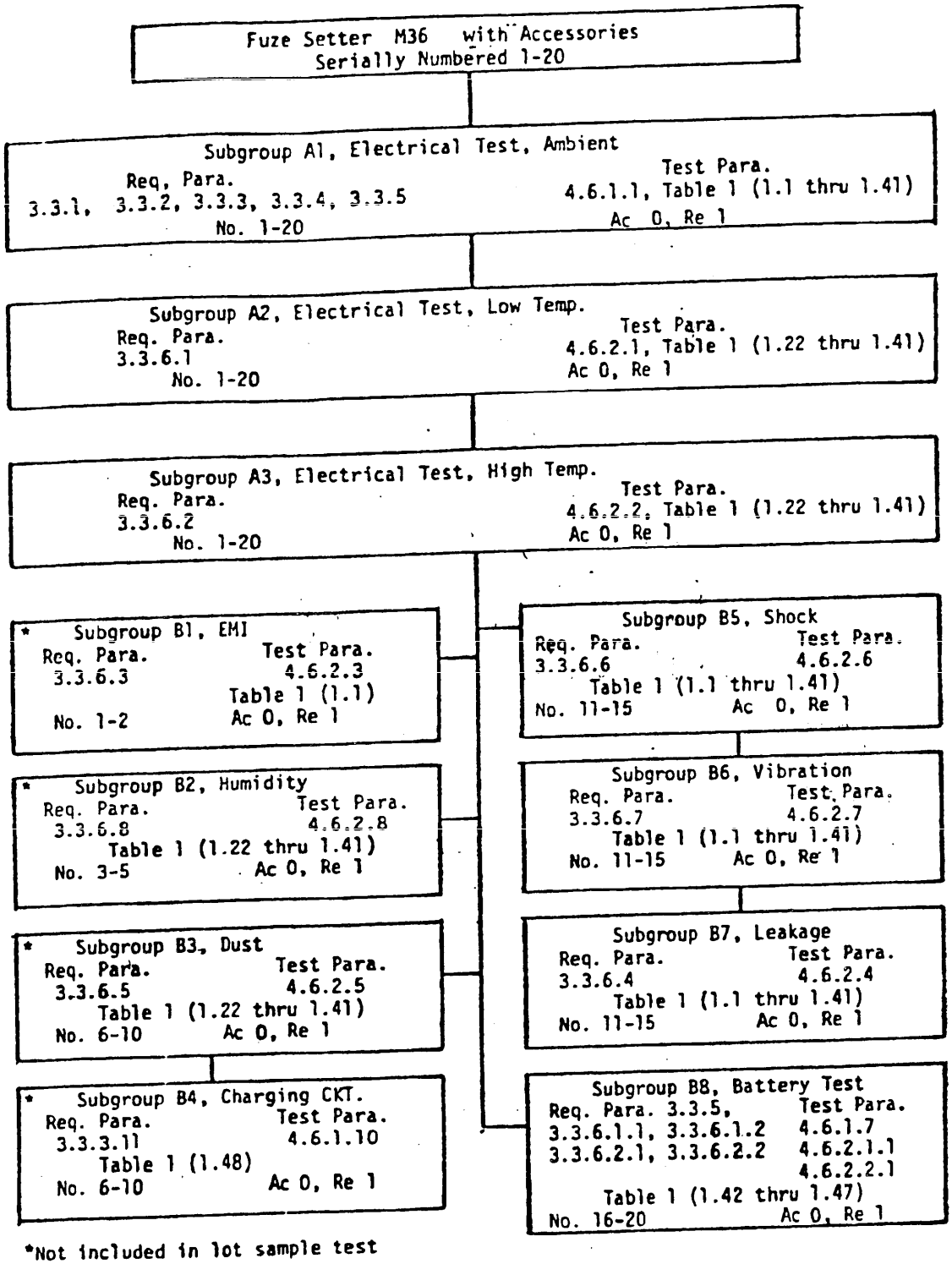
TABLE 2

The following table describes how the tests in Table 1 are performed using the block diagram of Figure 2.

Test No.	Switches S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	Remarks
1.1-1.21	-[1]	-	-	-	-	X[2]	1	-	-	X	read display
1.22	-	-	X	-	-	X	1	-	-	X	read display
1.23	-	-	-	X	-	X	1	-	-	X	read display
1.24	-	-	X	X	-	X	1	-	-	X	read display
1.25	-	X	-	-	-	X	1	-	-	X	read display
1.26	-	-	-	-	-	X	1	-	-	X	read display
1.27	-	-	-	-	-	X	1	-	-	X	read display
1.28	-	-	-	-	-	X	1	-	-	X	read display
1.29	-	-	-	-	-	-	2	-	-	X	read fuze time counter
1.30	-	-	-	-	-	X	1	X	-	X	read display
1.31	-	-	-	-	-	X	1	-	X	X	read display
1.32	-	-	-	-	-	X	1	-	-	-	read display
1.33	-	-	-	-	-	X	1	-	-	X	read turn-on time counter
1.34	-	-	-	-	-	X	1	-	-	X	initial condition
	-	-	-	-	-	-	1	-	-	X	read turn-off time counter
1.35	-	-	-	-	-	X	1	-	-	X	read period avg. counter
1.36	-	-	-	-	-	X	1	-	-	X	read +Vp on DVM
1.37	-	-	-	-	-	X	1	-	-	X	read -Vp on DVM
1.38	-	-	-	-	-	X	1	-	-	X	read -Vx on DVM
1.39			N/A								
1.40	-	-	-	-	-	X	1	-	-	X	read duration of +Vp on counter
1.41	-	-	-	-	-	X	1	-	-	X	read duration of -Vp on counter
1.42	X	-	-	-	-	-	1	-	-	X	initial condition
	-	-	-	-	X	-	1	-	-	X	read event counter
1.43	-	-	-	-	X	-	1	-	-	X	read event counter
1.44	X	-	-	-	-	-	1	-	-	X	initial condition
	-	-	-	-	X	-	1	-	-	X	read event counter
1.45	-	-	-	-	X	-	1	-	-	X	read event counter
1.46	X	-	-	-	-	-	1	-	-	X	initial condition
	-	-	-	-	X	-	1	-	-	X	read event counter
1.47	-	-	-	-	X	-	1	-	-	X	read event counter
1.48	X	-	-	-	-	-	1	-	-	X	read ammeter

[1] indicates open switch
[2] indicates closed switch

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*Not included in lot sample test

Figure 1. FAAS and Lot Sample Flow Chart

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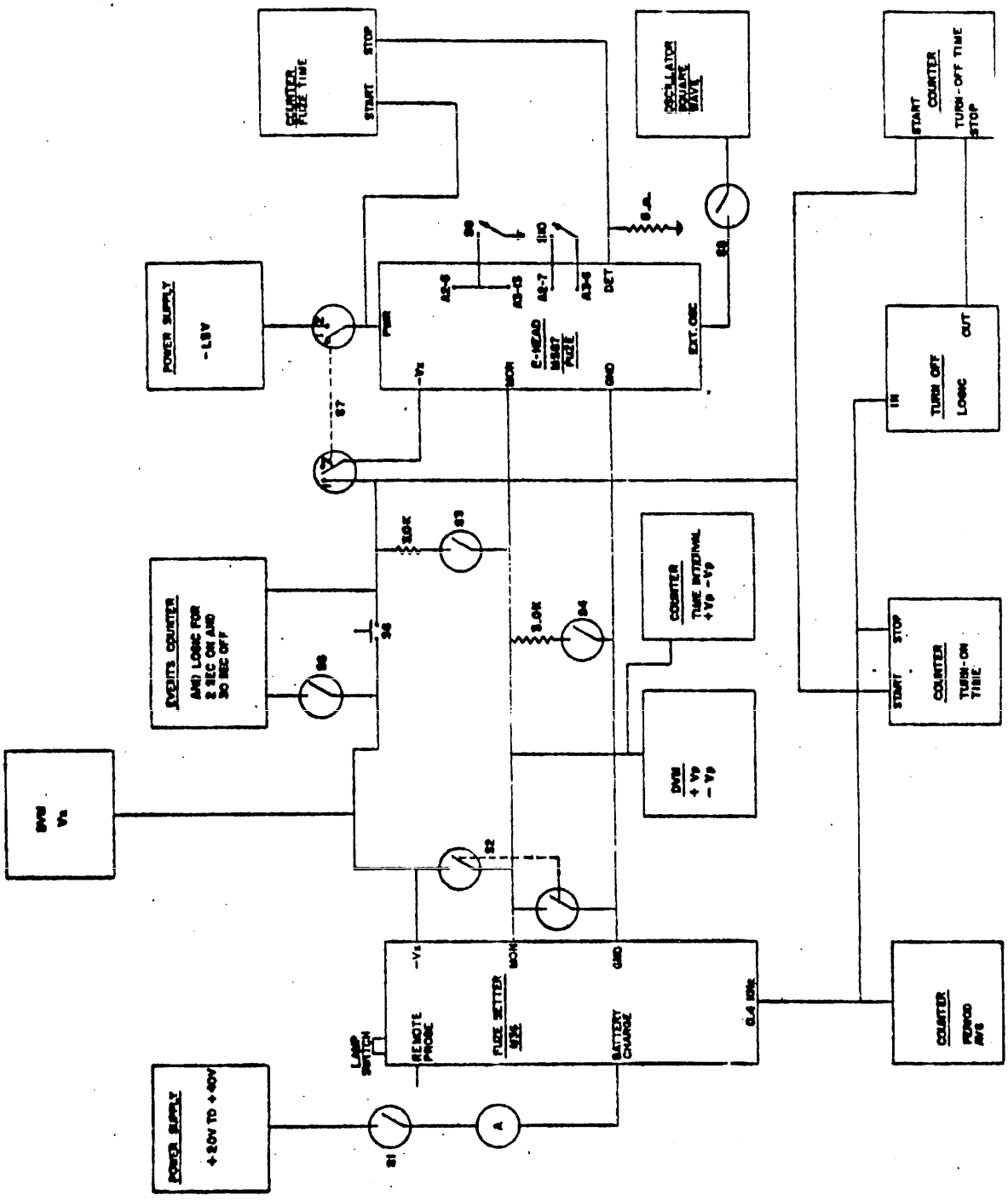


FIGURE 2. BLOCK DIAGRAM OF FUZE SETTER TESTER
(SEE TABLE 2 AND PARAGRAPH 4.5.2.1)

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